

# Stage 0 – Student Setup Instructions

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**Objective:** Prepare your computer with Python, Miniconda, Git, and the Bootcamp repository so you are ready for class.

Note: You will also install Visual Studio Code (VS Code) for editing code and running notebooks.

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## Part 1 – Install Miniconda

### 1. Download Miniconda

- Go to: <https://docs.conda.io/en/latest/miniconda.html>
- Choose the installer for your operating system:
  - **Windows:** Miniconda3 Windows 64-bit installer (.exe)
  - **Mac:** Miniconda3 macOS Intel or ARM (M1/M2) installer (.pkg or .sh)
  - **Linux:** Miniconda3 Linux 64-bit installer (.sh)

### 2. Run the Installer

- **Windows:** Double-click the **.exe** file, choose "Just Me", accept defaults, and let it add Miniconda to your PATH.
- **Mac/Linux:** Open a terminal and run:

```
bash Miniconda3-latest-<your-os>.sh
```

Follow the prompts (press **Enter** to accept defaults).

### 3. Verify Installation

- Close and reopen your terminal or Anaconda Prompt.
- Run:

```
conda --version
```

You should see something like **conda 24.x.x**.

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## Part 2 – Create Your Virtual Environment

### 1. Create environment named **bootcamp\_env**

```
conda create -n bootcamp_env python=3.10
```

## 2. Activate the environment

```
conda activate bootcamp_env
```

## 3. Install required packages

```
pip install jupyterlab python-dotenv numpy
```

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# Part 3 – Test with JupyterLab

## 1. Launch JupyterLab

```
jupyter lab
```

This will open JupyterLab in your web browser.

## 2. Create a new Python notebook

- Click **Python [conda env:bootcamp\_env]** as the kernel.
- In the first cell, type:

```
print("Hello, Bootcamp!")
```

- Run the cell (Shift + Enter). You should see:

```
Hello, Bootcamp!
```

## 3. Save the notebook

- Name it `python_tutorial`
  - It will save as `python_tutorial.ipynb`
  - Close JupyterLab in the browser.
  - Go to command window and press ctrl-c
  - if you are done with python, run `conda deactivate`
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# Part 4 – Install Git

## 1. Check if Git is installed

```
git --version
```

If you see a version number, skip to Part 5.

## 2. Install Git

- **Windows:** Download from <https://git-scm.com/download/win> and install with defaults.
- **Mac:**

```
brew install git
```

(Requires Homebrew — install from <https://brew.sh> if missing)

- **Linux:**

```
sudo apt-get update && sudo apt-get install git
```

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## Part 5 – Configure Git

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

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## Part 6 – Create Your Bootcamp Repository

### 1. On GitHub

- If you do not already have one, sign up for a GitHub account.
- Sign in to your GitHub account.
- Click **New Repository**.
- Name it:

```
bootcamp_<firstname>_<lastname>
```

Example: `bootcamp_jane_doe`

- Keep it **public** (unless instructed otherwise).

- Check **Add a README file**.
- Click **Create repository**.

## 2. Clone the repository locally

```
git clone https://github.com/<your-username>/bootcamp_<firstname>_<lastname>.git
cd bootcamp_<firstname>_<lastname>
```

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## Part 7 – Set Up Folder Structure

```
mkdir homework
mkdir project
mkdir class_materials
mkdir homework/homework0
```

### Class materials you receive during the course

Course files are shared one at a time. For each stage, make **one folder** inside `class_materials/` named exactly as the stage, and put **all** of that stage's files in it — reading, lecture notebook, homework sheet, homework starter, and any data files:

```
mkdir class_materials/stage03_python-fundamentals
```

Why this matters:

- Some lecture notebooks read a data file that must sit in the **same stage folder**. Stage 03's notebook loads `data/raw/starter_data.csv`, so that CSV has to be inside `class_materials/stage03_python-fundamentals/data/raw/`.
- **Run lecture notebooks from that folder**. They create any folders they need beside themselves. Running one from `Downloads` will scatter files there instead.
- **Never edit a file in `class_materials/`**. To do the homework, copy the starter into `homework/homeworkN/` first and work on the copy, so you always keep a clean original.
- `class_materials/` is never pushed to GitHub (see Part 8), so nothing you keep here affects your submissions.

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## Part 8 – Create `.gitignore`

Create a `.gitignore` file in the root of your repo:

```
# Ignore local class materials
class_materials/

# Ignore Python cache files
__pycache__/
*.pyc
*.pyo

# Ignore Jupyter checkpoints
.ipynb_checkpoints/

# Ignore environment files
.env

# Ignore OS files
.DS_Store
```

## Part 9 – Update README.md

Open `README.md` and replace with:

```
# Bootcamp Repository

## Folder Structure
- **homework/** → All homework contributions will be submitted here.
- **project/** → All project contributions will be submitted here.
- **class_materials/** → Local storage for class materials. Never pushed to GitHub.

## Homework Folder Rules
- Each homework will be in its own subfolder (`homework0`, `homework1`, etc.)
- Include all required files for grading.

## Class Materials Rules
- Each stage's handouts go in their own subfolder, named exactly as the course folder, e.g. `class_materials/stage01_problem-framing-and-scoping/`.
- Run lecture notebooks in place from that folder.
- Copy a homework starter into `homework/homeworkN/` before working on it.

## Project Folder Rules
- Keep project files organized and clearly named.
- The project folder structure is set up in Stage 02.
```

## Part 10 – Add the Notebook

Move the Python tutorial notebook into `homework/homework0/`:

```
mv /path/to/python_tutorial.ipynb homework/homework0/
```

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## Part 11 – Commit and Push

```
git add .  
git commit -m "Initial repo setup with homework0 notebook"  
git push origin main
```

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## Part 12 – Install Visual Studio Code (VS Code)

VS Code will make editing Python files.

### 1. Download VS Code

- Go to: <https://code.visualstudio.com/Download>
- Choose the installer for your OS and download it.

### 2. Install VS Code

- Run the installer and accept defaults.
- On Windows, check **Add to PATH** and **Add “Open with Code” to context menu** during setup.
- On Mac, drag the app into your **Applications** folder.

 You are now ready for class!